Technical Screening Test

Background

IDB publications are an essential means for sharing knowledge about development with governments, academics, and citizens of Latin America and the Caribbean, as well as around the world. Their visibility on the internet and discoverability in digital libraries largely depends on high-quality metadata, both in terms of alignment to metadata standards as well as the metadata values assigned to describe a publication and its content, including topics, countries, keywords, etc. Inconsistent, inaccurate, or missing metadata directly reduces the impact of our research.

The Challenge

Using the JSON-LD files provided, perform a programmatic analysis of the full text of the selected working papers to audit and improve our current metadata.

Files:

- **working_papers_metadata.json:** Contains 150 working papers, including structured metadata aligned with the Schema.org vocabulary, linked taxonomy terms, and the full text content of each document. The working papers may be written in English, Spanish, French or Portuguese.
- **taxonomy_labels.json:** Contains the taxonomy terms referenced by their URI in the working papers metadata, including their multilingual labels (English, Spanish, French, and Portuguese) and concept scheme information.

Your objective is threefold:

1. **Metadata Accuracy**
    - Determine if assigned taxonomy terms accurately reflect the document content (its *"aboutness"*) based on objective criteria that you define. Specify the defined criteria and provide a justification. Suggest terms that should be deleted.
    - Detect key concepts or entities present in the text but missing in the original metadata. Suggest new terms that should be added. Provide a rationale.
2. **Metadata Consistency**
    - Detect missing values in descriptive metadata fields, as well as values that are incorrect, or that do not correspond to a field. Suggest changes that should be made. Feel free to suggest the addition of new metadata fields, or the removal of existing fields, if you consider that appropriate. Provide a rationale.
3. **Optimization Proposal**

- Propose a scalable process or algorithm to automate and improve metadata assignments for the current stock of publications, as well as the flow of all future publications.

Deliverables

To be considered for the next round of the selection process, **please submit the following by midnight on Monday, March 1st**:

- **Code**: Clean, well-documented, and reproducible code that implements your analytical approach and outputs a **Comparative Metadata Dataset** with supporting evidence. Please include:
  - A brief README explaining how to run your solution.
  - A requirements.txt or environment file listing required dependencies.

  Python is preferable, but not a requirement
- **Comparative Metadata Dataset**: A CSV or JSON file comparing the original metadata with your suggested changes for all 150 documents. Include:
  - An "accuracy score" (or equivalent metric you define) for each concept or entity identified.
  - Suggested term additions and removals.
  - Flags for missing or inconsistent descriptive metadata fields.
- **Technical Report:** A brief document (PDF/Markdown) explaining:
  - Your methodology (e.g., why you chose specific NLP techniques).
  - Your criteria (e.g., how you concretely defined "Accuracy", etc.).
  - Key findings: A clear narrative summary of systemic issues found. Include tables and charts as necessary to support the findings.
  - Your proposal for scaling this analysis to the 13,000+ publications in the IDB Catalog.

Note: A core component of this test is your ability to define a technical framework for "Accuracy." We are not looking for just a review, but also for a reproducible logic. In your report, explicitly define your criteria for why metadata is considered "Correct," "Incorrect," or "Missing", etc.